

Solution Requirements

Date	26 June 2026
Team ID	SWTID-2026-4231
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Secure user access via email/password or SSO integration for policy analysts.</i>	High
2	Authorization levels	<i>Two roles: 'Admin' (can update ML models) and 'User' (can run predictions).</i>	High
3	External interfaces	<i>Integration with UNDP/World Bank Open Data APIs for automated dataset fetching.</i>	Medium

4	Transactions processing	<i>Handling input of numerical socio-economic data and outputting a predicted HDI score.</i>	High
5	Reporting	<i>Dashboard generation showing HDI score, visual trend charts, and export to PDF/CSV.</i>	High
6	Business rules, etc.	<i>System must validate that all 4 input fields are within historically realistic ranges (e.g., GNI > 0).</i>	High
7	Compliance to laws or regulations	<i>Adherence to data privacy standards (GDPR) regarding the storage of user input queries.</i>	High
8	<i>Other</i>	<i>Language toggle (English/Regional) for global accessibility.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>API response time for HDI prediction must be instantaneous.</i>	<i>Loads in < 2 seconds</i>

2	Scalability	<i>System should handle multiple users running simultaneous simulations.</i>	<i>Supports 500+ concurrent users</i>
3	Security & Data Privacy	<i>All data transmitted between the web UI and Flask API must be secure.</i>	<i>HTTPS/TLS 1.2+ encryption</i>
4	Reliability & Availability	<i>The system hosted on Render must be available consistently for daily analysis.</i>	<i>99.5% Uptime</i>
5	Usability & Accessibility	<i>Intuitive dashboard design using simple sliders and clear input labels.</i>	<i>WCAG 2.1 AA Compliance</i>
6	Other	<i>Codebase must be modular and documented for easy updates to ML models.</i>	<i>Maintainable/Scalable</i>